Standard Operating Procedure (SOP) for Smart Robotic Arm System

Clinical Operation Protocol

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Recognition Scope and System Initiation

1.1 Categories of Recognizable Medical Items

The system is designed to identify and classify the following four major categories of medical items within the surgical field:

- **Surgical Instruments:** Forceps, scissors, needle holders, retractors, etc.
- **Sharps:** Suture needles, scalpel blades, injection needles, etc.
- **Surgical Dressings:** Gauze, lap sponges, cotton balls, etc.
- **Miscellaneous Items:** Suction tubes, electrocautery tips, and other medical materials.

1.2 Task Initiation via Natural Language

Tasks are driven by natural language commands (voice or text).

- **Standard Commands:** Operators should use everyday clinical terminology.
- **Example:** "Count all scissors and forceps on the table," or "Locate the gauze on the far right."
- **Constraints:** Personnel are not required to provide coordinates, object labels, or pre-recorded quantities. The system is strictly prohibited from describing the background environment (e.g., tables, computers).

1.3 Basic Counting Principles

The system follows a "fixed counting sequence": top-to-bottom, left-to-right, or small-to-large. Random jumping is prohibited to ensure consistency and dual confirmation.

Exception Handling and Active De-cluttering

In clinical settings, instrument overlapping or occlusion is a common phenomenon. When the system detects uncertainty, guessing the quantity is strictly prohibited.

2.1 Trigger Conditions

If any of the following situations occur, the system will interrupt the routine count:

- Severe overlapping or stacking of instruments.
- Visual occlusion occurs, preventing the confirmation of the exact quantity with high confidence.

2.2 Active De-cluttering Procedure

When stacking uncertainty is detected, the system must execute the following steps:

1. **Announcement:** The system issues a text/voice prompt: "Initiating active de-cluttering," accompanied by the target 2D coordinates $[(x, y)]$.
2. **Robotic Action:** The robotic arm is automatically driven to the specified area to perform "scattering" or "spreading" actions.
3. **Verification:** After the arm retracts, the visual system rescans and recognizes the area to update the count.

2.3 Fallback Mechanism

If automated de-cluttering cannot resolve the uncertainty, the system will escalate the task by requesting manual intervention from the nursing staff.

Human-Robot Collaboration and Safety Protection

3.1 Human Proximity Monitoring

The system continuously monitors the distance $d(t)$ between the robotic arm end-effector $p_{ee}(t)$ and the personnel $p_h(t)$:

$$d(t) = \|p_{ee}(t) - p_h(t)\|$$

3.2 Safe Pause and Autonomous Recovery

- **Safe Pause:** If $d(t)$ falls below the safety threshold, the system enters "Safe Pause Mode."
- **Autonomous Recovery Conditions:** The system will only resume the task when the following conditions are met simultaneously:
    1. **Spatial Condition:** The human-robot distance is greater than or equal to the safe distance ($d(t) \geq d_{safe}$).
    2. **Temporal Condition:** The safe distance has been continuously maintained for a specified period of time.

Digital Records and Traceability

Every counting task must generate a digital record for cross-temporal traceability.

- **Mandatory Elements:** Timestamp, ROI (Region of Interest) image snapshot, instrument type, and quantity.